

How to learn the talk with this booklet

The booklet has three parts, each with one page per slide. Work through them in order, slide by slide, always **aloud**, standing if you can.

1. **Learn the skeleton, not the words.** Per slide, learn verbatim only the *opening sentence*, the *last sentence* (it is the bridge to the next slide) and the *numbers*. Everything in between is a list of points you say in your own words. A talk recited word for word sounds recited and collapses on the first slip; anchor sentences give you a safe place to land.
2. **Part A – full text.** One sentence per line; every line break is a pause. Read the slide twice, then cover the text and say it while looking at the thumbnail.
3. **Part B – first letters.** Say the slide from the skeleton. Peek only when stuck. Move on once you manage two clean runs.
4. **Part C – anchors only.** Say the whole slide from the opening and closing sentence alone. This is the level you need on stage.
5. **Time it.** Each page carries its time window. The pauses are part of the timing. Target for the whole talk: **9:30**.
6. **Space it out.** Day 1: slides 1–5 to Part C. Day 2: slides 6–11, then re-test 1–5. Day 3: two full runs with the PDF full-screen and a clicker, one of them recorded on your phone – listen back. Day 4: open the booklet on a random page and start from there (this trains recovery from a blank), plus the questions in QA.md. Last evening: one relaxed run. **Freeze the script two days before the talk;** every edit resets what you have learnt.
7. **Anki.** `anki.csv` holds one card per slide, one card per bridge sentence and one per number – for the minutes between sessions.

Legend. **[CLICK]** – advance the slide. *grey italics* – **[CUT IF LATE]**: can be dropped without breaking the argument. Numbers table on the last page.



Good morning.

I'm Martin Maxa from the Department of Measurement, here at this faculty.

This is joint work with Jakub Svatoš from the same department.

Today I'll talk about detecting, classifying and localizing impulsive sounds – in our case, gunshots.

The main part is gunshot classification, where we compare different cepstral features and two classifiers.

And I'll briefly show how the system localizes the source.

Motivation and research questions

- Acoustic detection of gunfire complements camera-based surveillance: the acoustic channel is omnidirectional and does not require line of sight.
- Three required functions, in outdoor environments that are noisy and reverberant:



detection
of the impulsive event



classification
ideally down to calibre



localization
of the source, in 2D

This study

1. Which cepstral representation separates gunshots from acoustically similar false alarms most reliably?
2. Neural network or support vector machine on those features?
3. How does the analysis frame length interact with environmental reflections?

Acoustic Detection of Impulsive Sounds · IMEKO TC4 2026

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This work is part of a larger project – an acoustic sensor system for protecting soft targets.

Acoustic detection is a good complement to cameras.

Sound is omnidirectional, and you don't need a line of sight.

[CUT IF LATE] *The full system has to detect an impulsive event, classify it – ideally down to the calibre – and localize the source.*

And it has to work outdoors, with noise and a lot of reflections. [CUT]

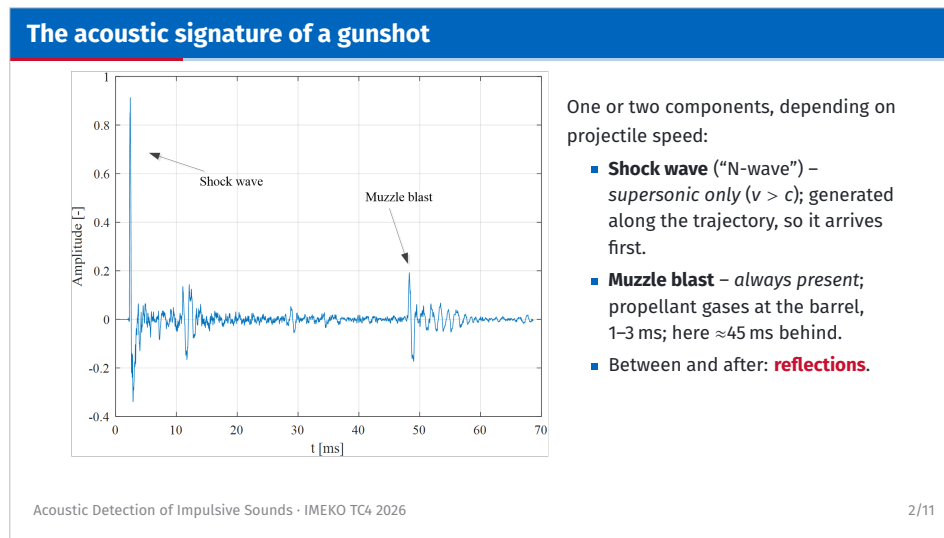
[CLICK] _____

In this study we ask three questions.

First, which cepstral features best separate gunshots from similar impulsive sounds?

Second, which classifier works better – a neural network or a support vector machine?

And third, how does the frame length affect the results?



Let's start with the signal of a gunshot.

A gunshot has one or two components, depending on the projectile speed.

A subsonic shot has only the muzzle blast.

A supersonic one also has a shock wave.

This recording is a supersonic shot, so we can see both.

The first one, at about two and a half milliseconds, is the shock wave.

[CUT IF LATE] *It's created along the bullet's path and has the typical N-shape.* [CUT]

[CLICK] _____

The second one, about forty-five milliseconds later, is the muzzle blast.

It comes from the gases leaving the barrel, and every gunshot has it.

Everything between and after them is mostly reflections from the surroundings.

This matters for the frame length, and I'll come back to it in the results.

Single-domain analysis is insufficient

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence
Effective feature extraction must integrate *both* time and frequency domain processing.

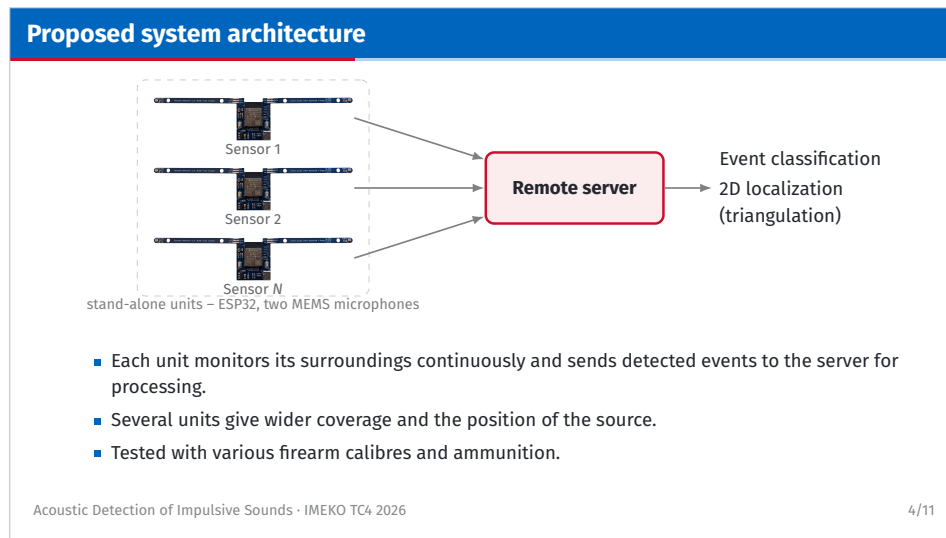
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This also shows why one domain alone is not enough.

The spectrum often tells you more about the environment than about the gunshot.

And the waveform changes as the sound travels and hits obstacles.

So we need features that combine both domains, and cepstral features do exactly that.



Now to the system itself.

It's a set of independent sensor units connected to a remote server.

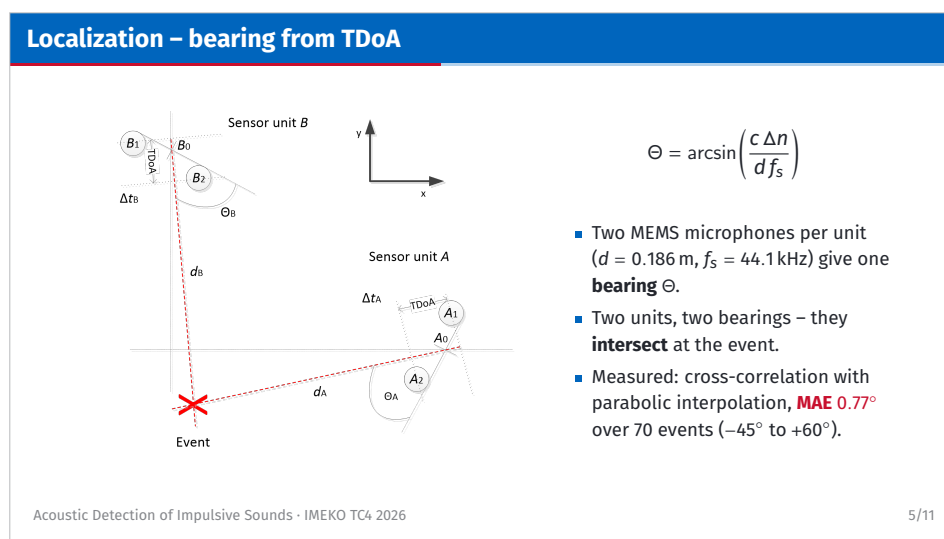
Each unit continuously monitors its surroundings.

When it detects an impulsive event, it sends the recorded signal to the server.

The server then does two things – it classifies the event and localizes the source.

More units mean wider coverage, and they let us estimate the position of the source.

We tested the system with several calibres and types of ammunition.



Let me briefly explain the localization first.

Each sensor unit has two MEMS microphones, about twenty centimetres apart, sampled at 44.1 kilohertz.

We measure the time difference of arrival between the two microphones. From this delay we get the direction to the source, using the equation here.

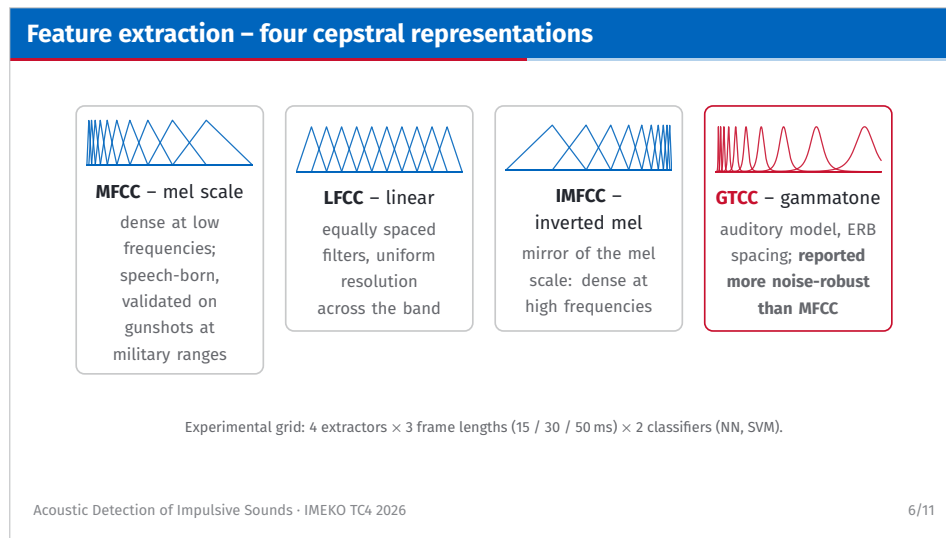
[CLICK] _____

So one unit gives us one direction.

With two units we get two directions, and where they cross is the position of the source.

We tested this on seventy events at nine reference angles.

With cross-correlation and parabolic interpolation, the mean absolute error was zero point seven seven degrees.



Now to the main part – classification.

We compared four cepstral feature extractors.

They differ in the filter bank they use.

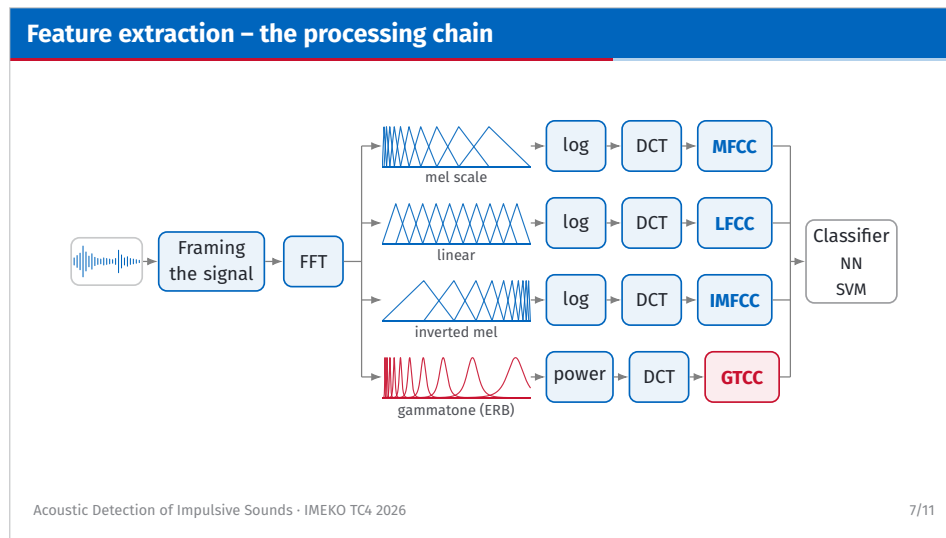
MFCC uses the mel scale, with dense filters at low frequencies.

It comes from speech processing, but it works for gunshots too.

LFCC has linearly spaced filters, and IMFCC uses an inverted mel scale, dense at high frequencies.

GTCC uses a gammatone filter bank, which is based on a model of human hearing.

It's also reported to be more robust to noise than MFCC.



And this is how the features are computed.

All four use the same chain – framing, FFT, the filter bank, then a logarithm and a DCT.

The filter bank is the only place where they differ, and here you can see all four of them on the same frequency axis.

The coefficients then go to a classifier.

We tested two – a neural network and a support vector machine – with three frame lengths, fifteen, thirty and fifty milliseconds.

Classification results – 9 mm class							
Frame	Extractor	ACC (%)		RLC (%)		MCC (–)	
		SVM	NN	SVM	NN	SVM	NN
15 ms	MFCC	83.90	85.59	80.65	67.74	0.62	0.62
	LFCC	83.90	83.90	80.65	64.52	0.62	0.57
	IMFCC	85.59	90.68	83.87	87.10	0.66	0.77
	GTCC	93.22	97.46	87.10	93.55	0.82	0.93
30 ms	MFCC	84.75	87.29	67.74	87.10	0.60	0.70
	LFCC	87.29	83.90	70.97	70.97	0.66	0.59
	IMFCC	85.59	92.37	67.74	80.65	0.62	0.80
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50 ms	MFCC	84.75	93.22	74.19	90.32	0.61	0.83
	LFCC	88.98	86.44	77.42	77.42	0.71	0.66
	IMFCC	89.83	90.68	74.19	80.65	0.73	0.76
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Classification results for the 9 mm class. Test partition: 118 events, 31 of them 9 mm.

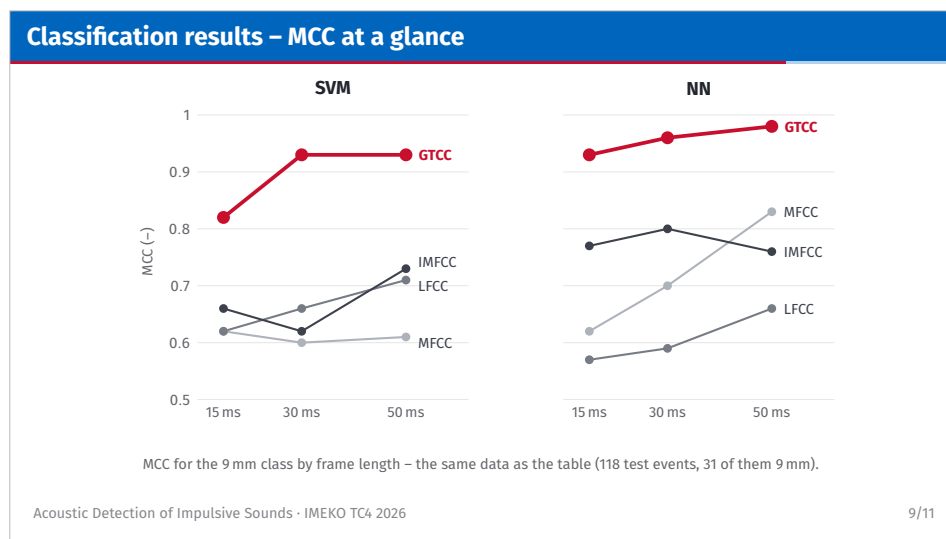
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This is the full table of results for the nine-millimetre class.

The GTCC rows are highlighted.

I won't go through the numbers – let's look at the same data as a chart.



Here are the same results, just the Matthews correlation coefficient.

MCC takes the whole confusion matrix into account, so it's less affected by class imbalance than accuracy.

The red line is GTCC.

It's on top in both panels, at every frame length.

The neural network does better than the SVM, and the gap grows with longer frames.

The best configuration is GTCC with the neural network and a fifty-millisecond frame.

That gives ninety-nine point one five percent accuracy, ninety-six point seven seven percent recall, and an MCC of zero point nine eight.

In other words, one hundred and seventeen correct decisions out of one hundred and eighteen.

Discussion

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** – better at modelling complex patterns, especially with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms – 99.15 % ACC, 96.77 % RLC, MCC 0.98.

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Three things to take from this.

First, MFCC, LFCC and IMFCC give similar results, so the filter spacing makes little difference.

Second, GTCC is clearly better.

[CUT IF LATE] *One likely reason is that the gammatone bank comes from a hearing model and is known to be more robust to noise.*

But we didn't test that directly. [CUT]

Third, the neural network is slightly better than the SVM, especially with longer frames.

That's a general trend, not true for every single result.

[CLICK] _____

And finally, longer frames generally improve accuracy.

But at fifty milliseconds, the frame already contains more and more reflections, not just the gunshot itself.

So even longer frames wouldn't necessarily help.

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you – questions?

svatoja1@fel.cvut.cz maxamart@fel.cvut.cz

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To sum up.

We presented a distributed acoustic sensor system that detects, classifies and localizes impulsive sounds.

For gunshot classification, cepstral features work well with both classifiers. GTCC gives the best results of the four.

And the neural network is slightly better than the SVM, especially with longer frames.

All of this was tested on real firearms, with several calibres and types of ammunition.

Thank you.



G m.

I'm M M f t D o M, h a t f.

T i j w w J S f t s d.

T I'll t a d, c a l i s - i o c, g.

T m p i g c, w w c d c f a t c.

A I'll b s h t s l t s.

Motivation and research questions

- Acoustic detection of gunfire complements camera-based surveillance: the acoustic channel is omnidirectional and does not require line of sight.
- Three required functions, in outdoor environments that are noisy and reverberant:



detection
of the impulsive event



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localization
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- Neural network or support vector machine on those features?
- How does the analysis frame length interact with environmental reflections?

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T w i p o a l p – a a s s f p s t.

A d i a g c t c.

S i o, a y don't n a l o s.

T f s h t d a i e, c i – i d t t c – a l t s.

A i h t w o, w n a a l o r.

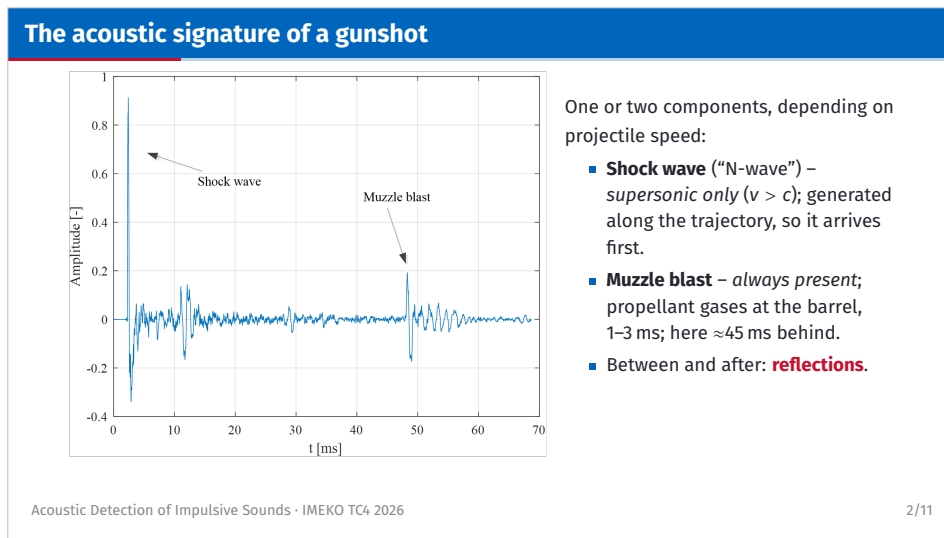
[CLICK]

I t s w a t q.

F, w c f b s g f s i s?

S, w c w b – a n n o a s v m?

A t, h d t f l a t r?



Let's s w t s o a g.

A g h o o t c, d o t p s.

A s s h o t m b.

A s o a h a s w.

T r i a s s, s w c s b.

T f o, a a t a a h m, i t s w.

It's c a t bullet's p a h t t N-s.

[CLICK] _____

T s o, a f-f m l, i t m b.

I c f t g l t b, a e g h i.

E b a a t i m r f t s.

T m f t f l, a I'll c b t i i t r.

Single-domain analysis is insufficient

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

Effective feature extraction must integrate *both* time and frequency domain processing.

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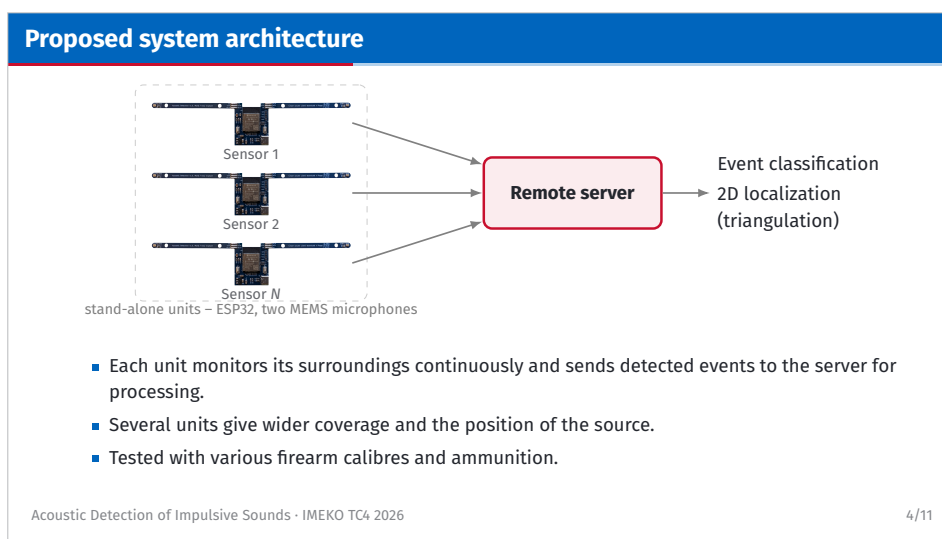
3/11

T a s w o d a i n e .

T s o t y m a t e t a t g .

A t w c a t s t a h o .

S w n f t c b d , a c f d e t .



N t t s i.

It's a s o i s u c t a r s.

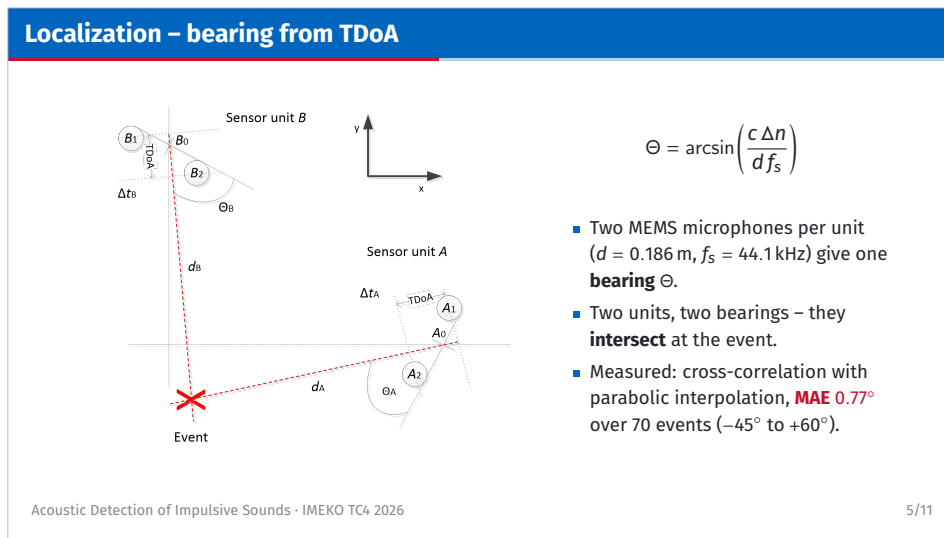
E u c m i s.

W i d a i e, i s t r s t t s.

T s t d t t – i c t e a l t s.

M u m w c, a t l u e t p o t s.

W t t s w s c a t o a.



L m b e t l f.

E s u h t MEMS m, a t c a, s a 44.1 k.

W m t t d o a b t t m.

F t d w g t d t t s, u t e h.

[CLICK]

S o u g u o d.

W t u w g t d, a w t c i t p o t s.

W t t o s e a n r a.

W c-c a p i, t m a e w z p s s d.

Feature extraction – four cepstral representations



MFCC – mel scale
dense at low frequencies;
speech-born, validated on gunshots at military ranges



LFCC – linear
equally spaced filters, uniform resolution across the band



IMFCC – inverted mel
mirror of the mel scale: dense at high frequencies



GTCC – gammatone
auditory model, ERB spacing; **reported more noise-robust than MFCC**

Experimental grid: 4 extractors × 3 frame lengths (15 / 30 / 50 ms) × 2 classifiers (NN, SVM).

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N t t m p – c.

W c f c f e.

T d i t f b t u.

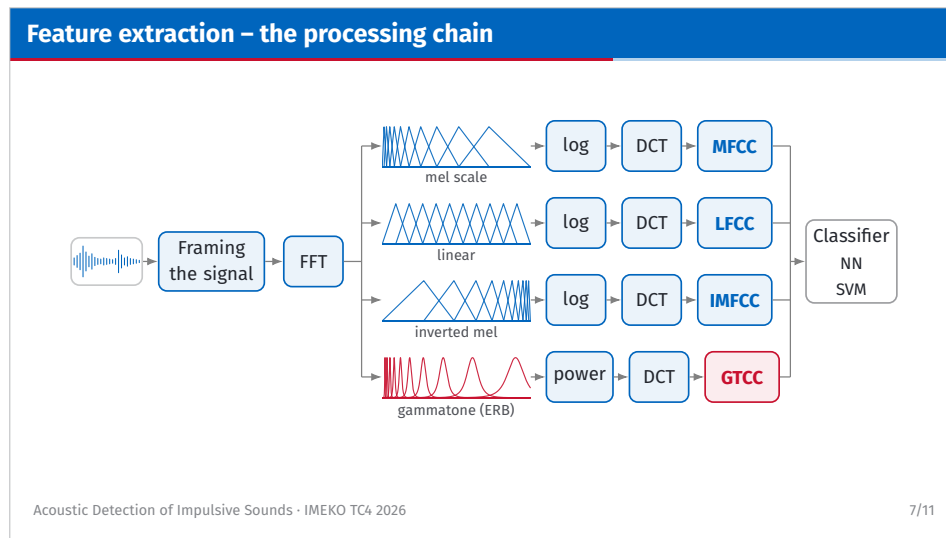
MFCC u t m s, w d f a l f.

I c f s p, b i w f g t.

LFCC h l s f, a IMFCC u a i m s, d a h f.

GTCC u a g f b, w i b o a m o h h.

It's a r t b m r t n t MFCC.



A t i h t f a c.

A f u t s c – f, FFT, t f b, t a l a a DCT.

T f b i t o p w t d, a h y c s a f o t o t s f a.

T c t g t a c.

W t t – a n n a a s v m – w t f l, f, t a f m.

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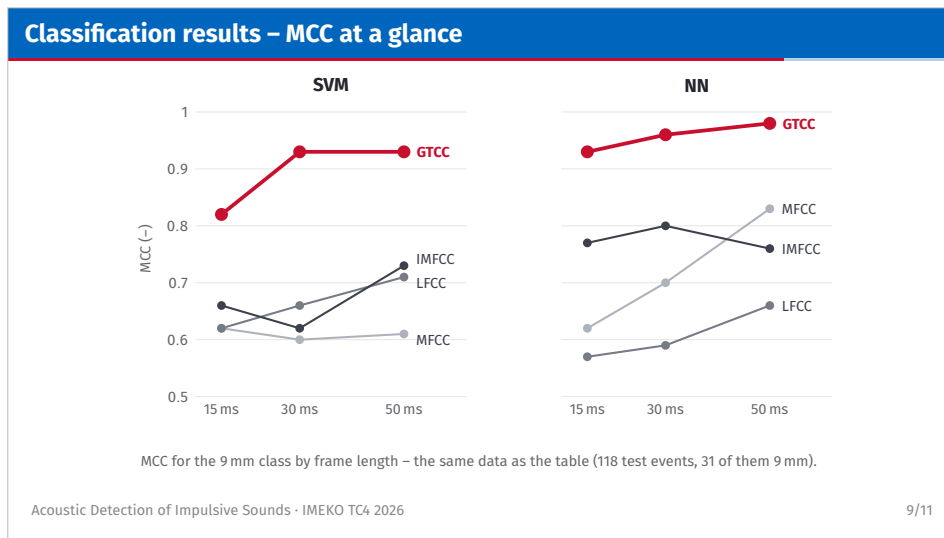
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T i t f t o r f t n-m c.

T GTCC r a h.

I won't g t t n – let's l a t s d a a c.



H a t s r, j t M c c.

MCC t t w c m i a, s i t's l a b c i t a.

T r l i GTCC.

I t's o t i b p, a e f l.

T n n d b t t SVM, a t g g w l f.

T b c i GTCC w t n n a a f-m f.

T g n-n p o f p a, n-s p s s p r, a a MCC o z p n e.

I o w, o h a s c d o o o h a e.

Discussion

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** – better at modelling complex patterns, especially with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms – 99.15 % ACC, 96.77 % RLC, MCC 0.98.

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T t t t f t.

F, MFCC, LFCC a IMFCC g s r, s t f s m l d.

S, GTCC i c b.

O l r i t t g b c f a h m a i k t b m r t n.

B w didn't t t d.

T, t n n i s b t t SVM, e w l f.

That's a g t, n t f e s r.

[CLICK] _____

A f, l f g i a.

B a f m, t f a c m a m r, n j t g i.

S e l f wouldn't n h.

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
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T s u.

W p a d a s s t d, c a l i s.

F g c, c f w w w b c.

GTCC g t b r o t f.

A t n n i s b t t SVM, e w l f.

A o t w t o r f, w s c a t o a.

T y.



Good morning.

...

And I'll briefly show how the system localizes the source.

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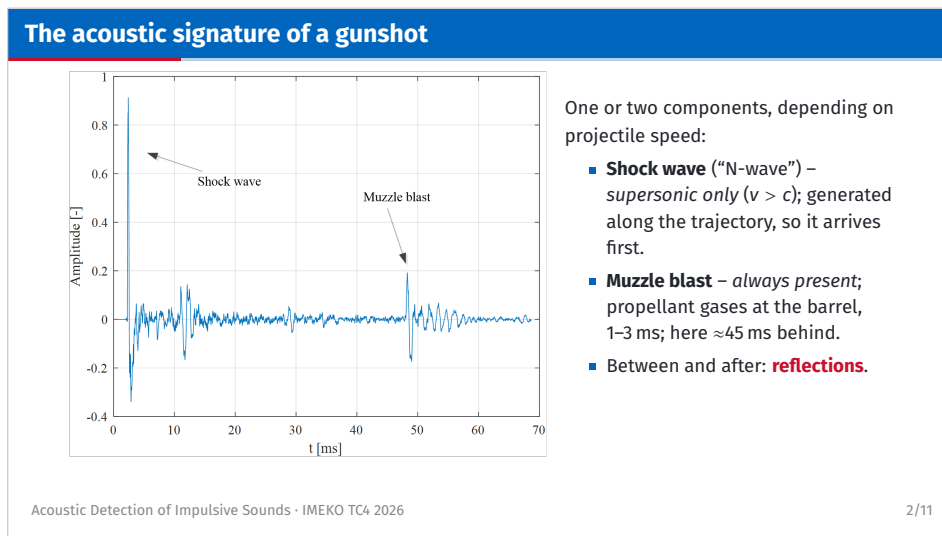
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This work is part of a larger project – an acoustic sensor system for protecting soft targets.

...

And third, how does the frame length affect the results?



Let's start with the signal of a gunshot.

...

This matters for the frame length, and I'll come back to it in the results.

Single-domain analysis is insufficient

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
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Consequence

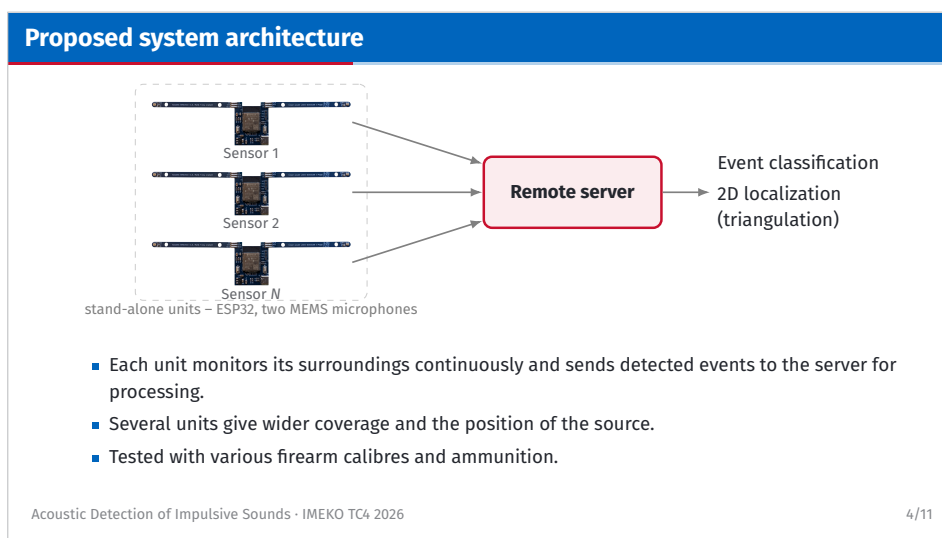
Effective feature extraction must integrate *both* time and frequency domain processing.

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• • •

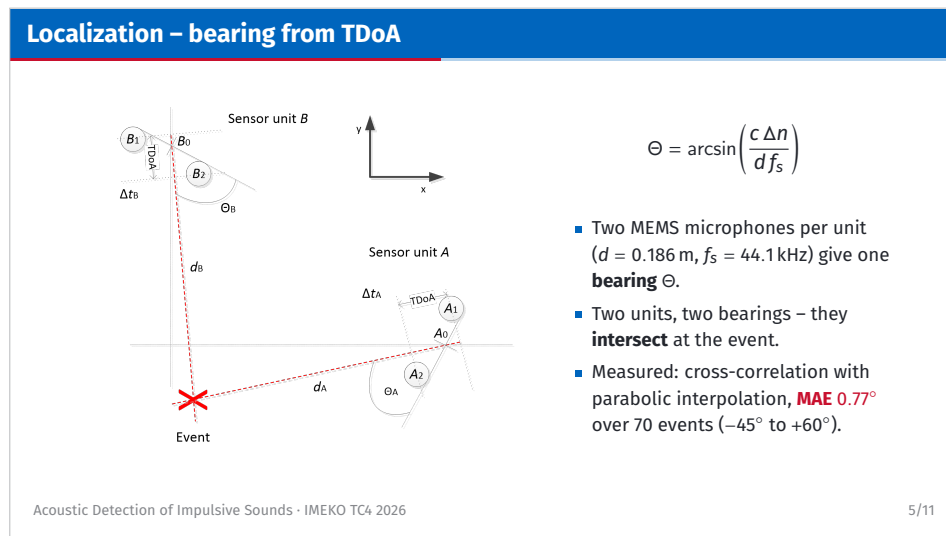
So we need features that combine both domains, and cepstral features do exactly that.



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We tested the system with several calibres and types of ammunition.



Let me briefly explain the localization first.

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With cross-correlation and parabolic interpolation, the mean absolute error was zero point seven seven degrees.

Feature extraction – four cepstral representations



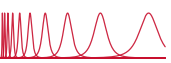
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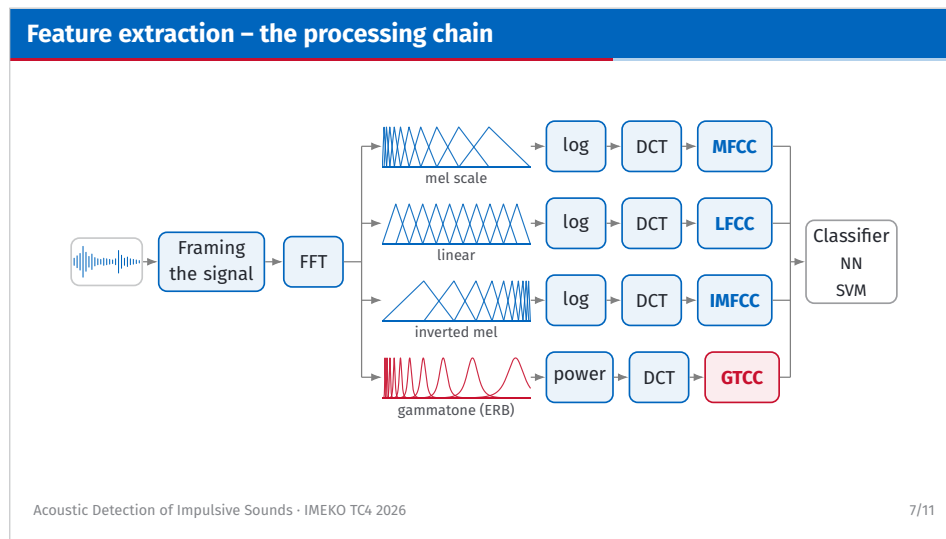
Experimental grid: 4 extractors × 3 frame lengths (15 / 30 / 50 ms) × 2 classifiers (NN, SVM).

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Now to the main part – classification.

...

It's also reported to be more robust to noise than MFCC.



And this is how the features are computed.

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We tested two – a neural network and a support vector machine – with three frame lengths, fifteen, thirty and fifty milliseconds.

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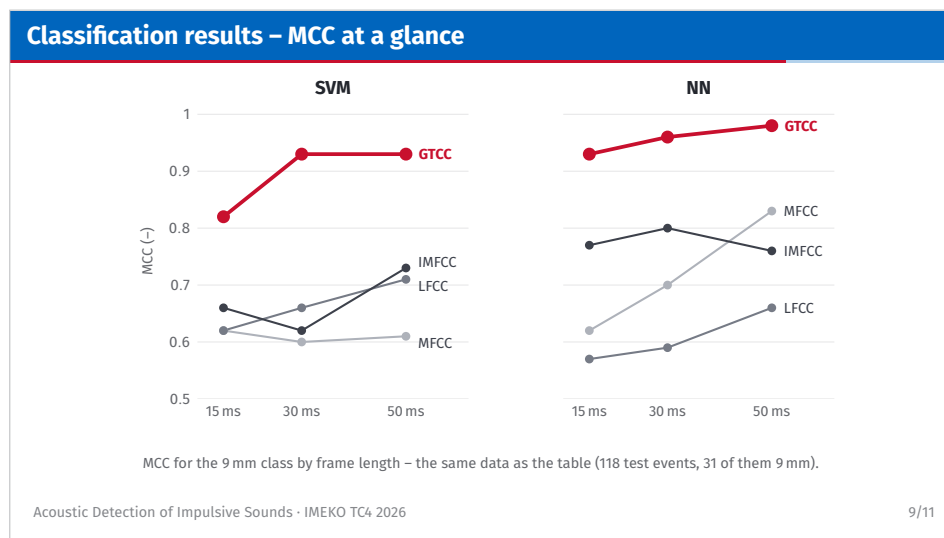
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This is the full table of results for the nine-millimetre class.

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In other words, one hundred and seventeen correct decisions out of one hundred and eighteen.

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- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** – better at modelling complex patterns, especially with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms – 99.15 % ACC, 96.77 % RLC, MCC 0.98.

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Three things to take from this.

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So even longer frames wouldn't necessarily help.

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you – questions?

svatoja1@fel.cvut.cz maxamart@fel.cvut.cz

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To sum up.

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Thank you.

Every number in the talk

Best configuration (GTCC + NN, 50 ms) – accuracy?	99.15 %
Best configuration (GTCC + NN, 50 ms) – recall?	96.77 %
Best configuration (GTCC + NN, 50 ms) – MCC?	0.98
Test partition – how many events, how many 9 mm?	118 events, 31 of them 9 mm
99.15 % accuracy = how many correct decisions?	117 of 118
96.77 % recall = how many 9 mm shots recalled?	30 of 31
Corpus – training / test samples (five classes)?	371 / 118
Features per extractor?	26 coefficients per frame
Frame lengths tested?	15 / 30 / 50 ms
Bearing accuracy (cross-correlation + parabolic interpolation)?	MAE 0.77°
Bearing evaluation – how many events, how many reference angles?	70 events, 9 angles (–45° to +60°)
Microphone spacing and sampling rate?	d = 0.186 m ($\approx$ 20 cm), f_s = 44.1 kHz
Maximum TDoA for this geometry?	$\approx$ 24 samples
Shock wave / muzzle blast timing in the recording shown?	shock wave $\approx$ 2.5 ms, muzzle blast $\approx$ 45 ms later, lasting 1–3 ms
Talk length – target?	10 min slot, script written to 9:30
